

→ Multicollinearity

But

income
↑
[w₁]

cs
↑
[w₂] → importance

f₁² f₂²
↘ ↗

$$f_1 = \alpha f_2 + \beta$$

→

Multicollinearity

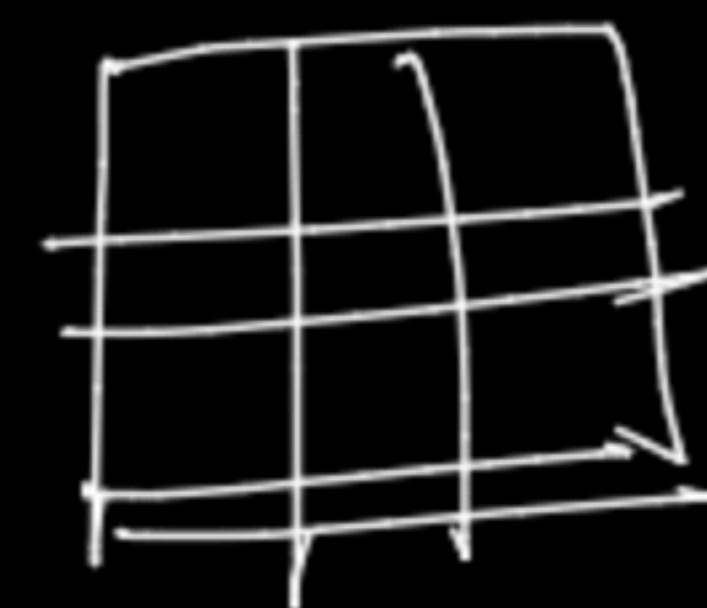
→

f₁ → f₂ f₃ f₄

$$f_1 \leftarrow f_2 + f_3 + f_4$$

→ [] → f(x)
↑
(FE)

(X, Y)



correlation matrix
↑ ↑

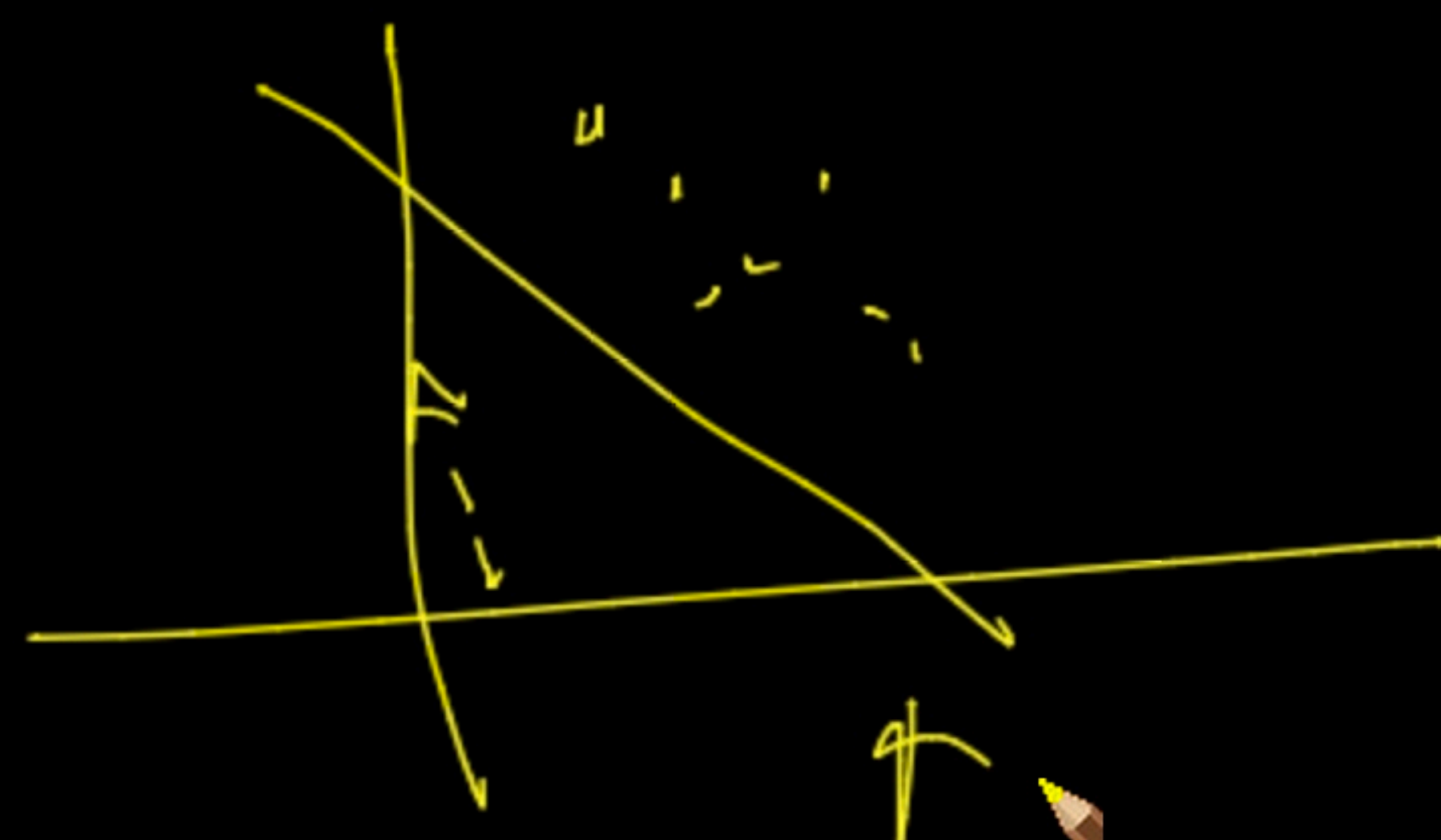
LR $\rightarrow$ linearly separable

$\nwarrow$
Non-linear

$\uparrow$
Feature eng

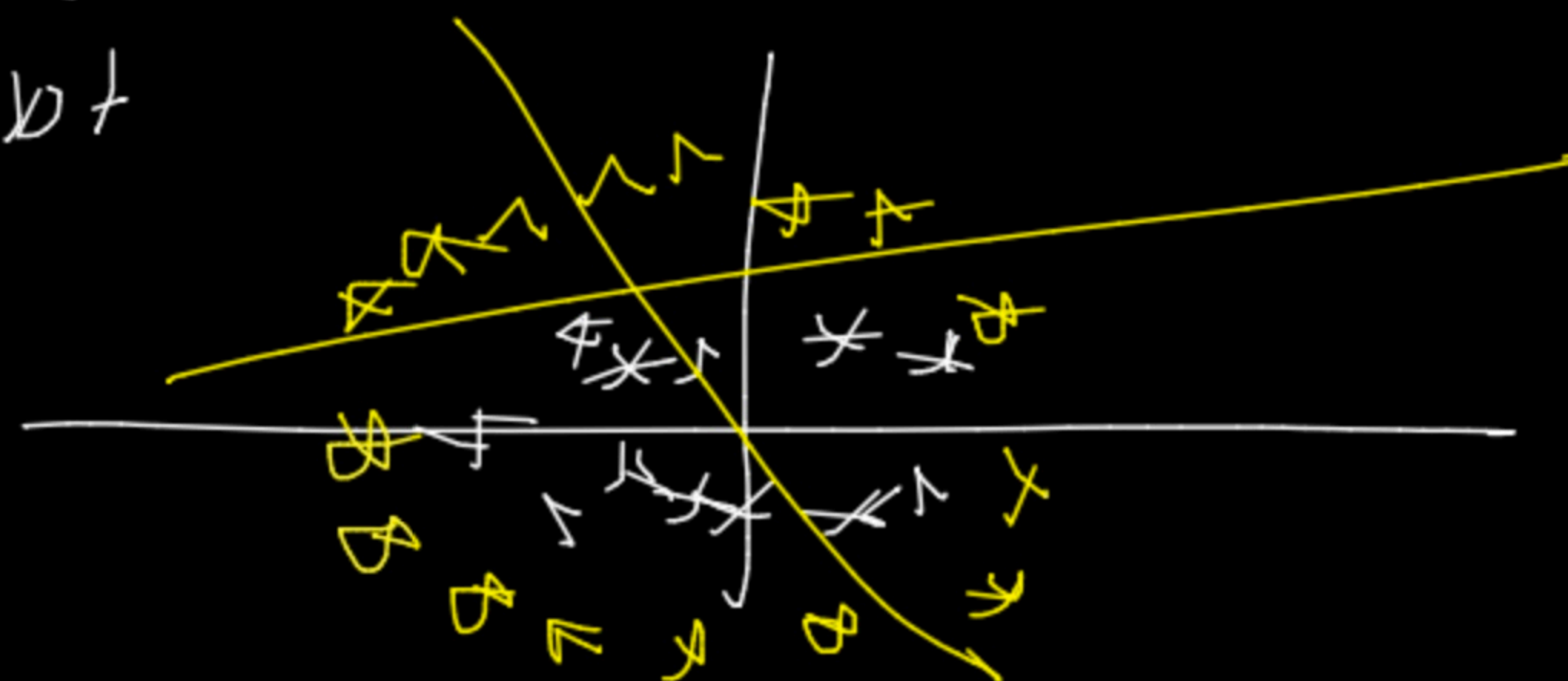
eg.

f_1	f_2	y
-	-	-
-	-	-
-	-	-
-	-	-



FE

$\nwarrow$
plot



Logistic Regression $\rightarrow$ classification

Linear Regression



Regression



$$D = (X, Y)$$



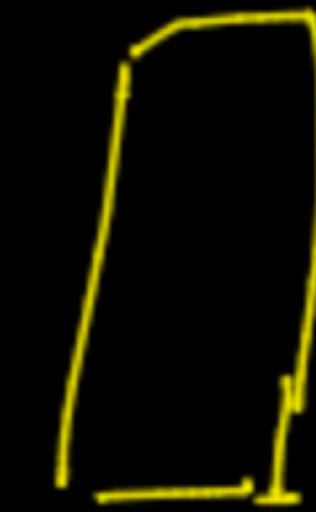
only Real No

eg.

hist

		X		Y
		Discount	BMI	
{	1	15	T	Sales Volume
	2	17	F	
	3	1	1	Sales Volume
	4	1	1	
	5	1	1	

$$LR(X, Y) \rightarrow$$



$\rightarrow ?$

per regression

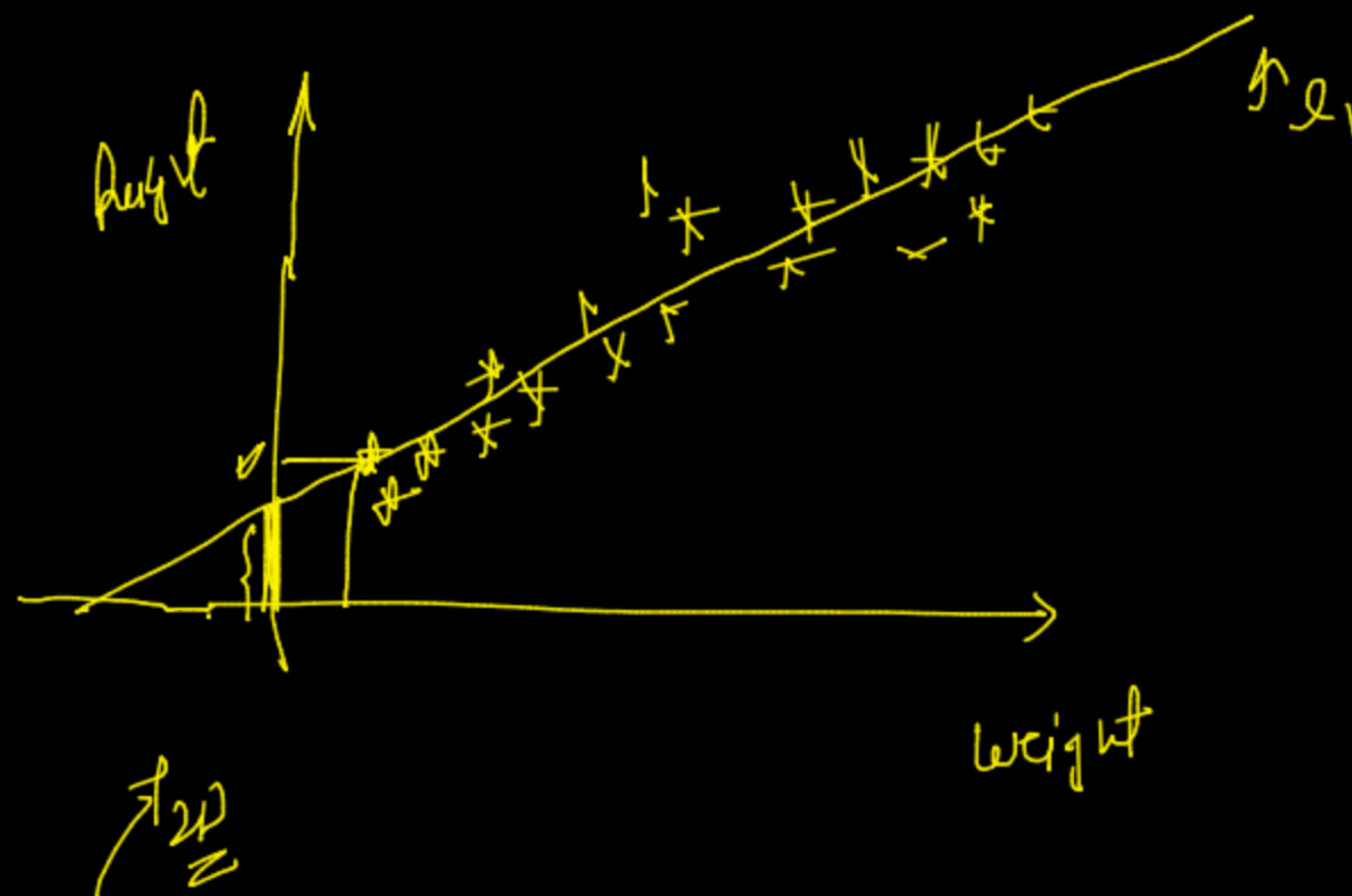
$\swarrow [15, T]$

Geometrically

weight, gender, ethnicity, hair-color $\rightarrow$ height?



weight $\rightarrow$ height



$$y = mx + c$$

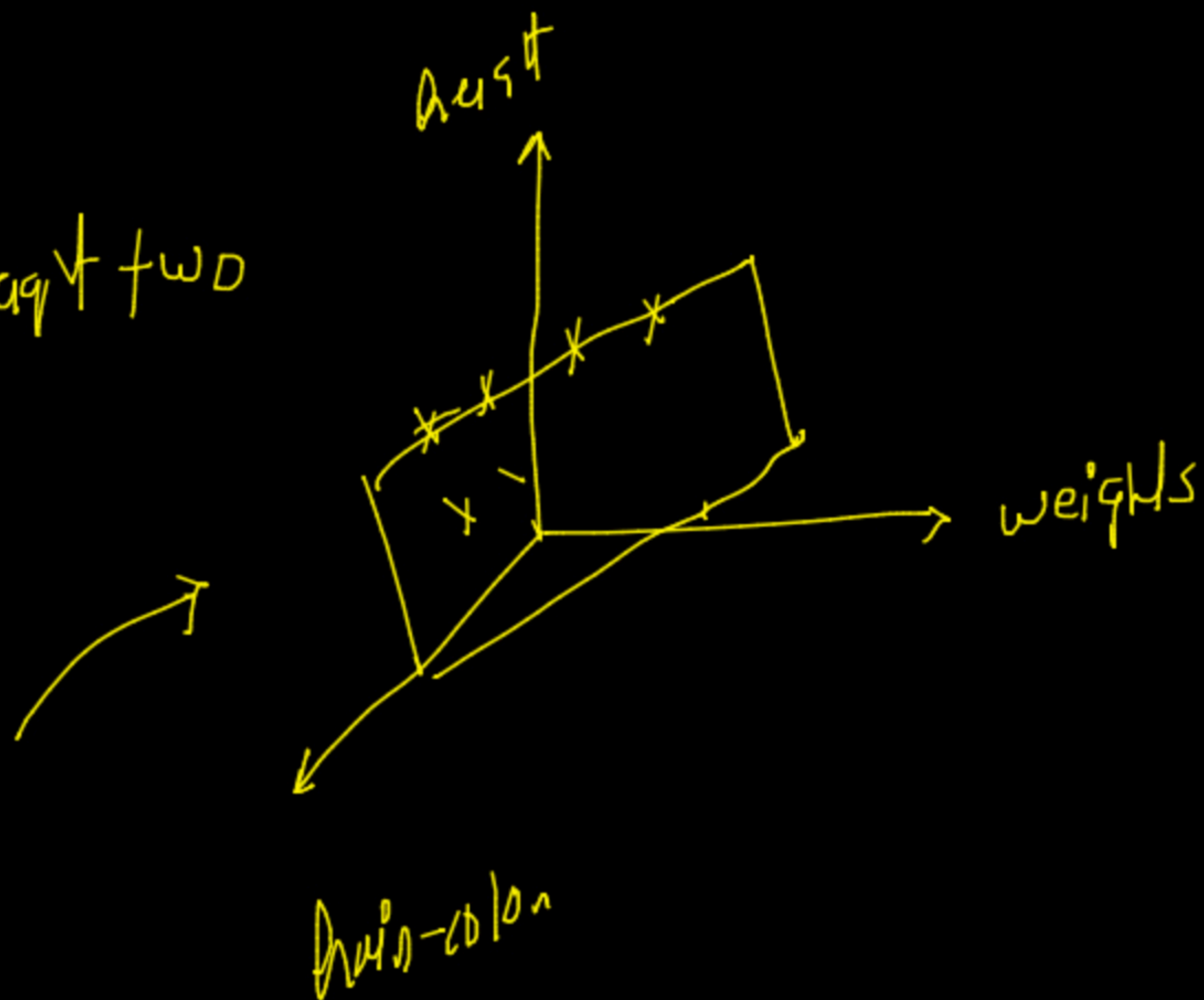
3D $\rightarrow$ Plane

Aim find (line) that fit the (training data)
 $\leftarrow$
 $\text{height} = w_1 * \text{weight} + w_0$

c_1
↑

$$\text{height} = w_1 \times \text{bias} + w_2 \times \text{weight} + w_0$$

↑ ↑



$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + \dots + w_n x_n + w_0$$

↑

$$y_i = w^T x_i + w_0 \rightarrow$$

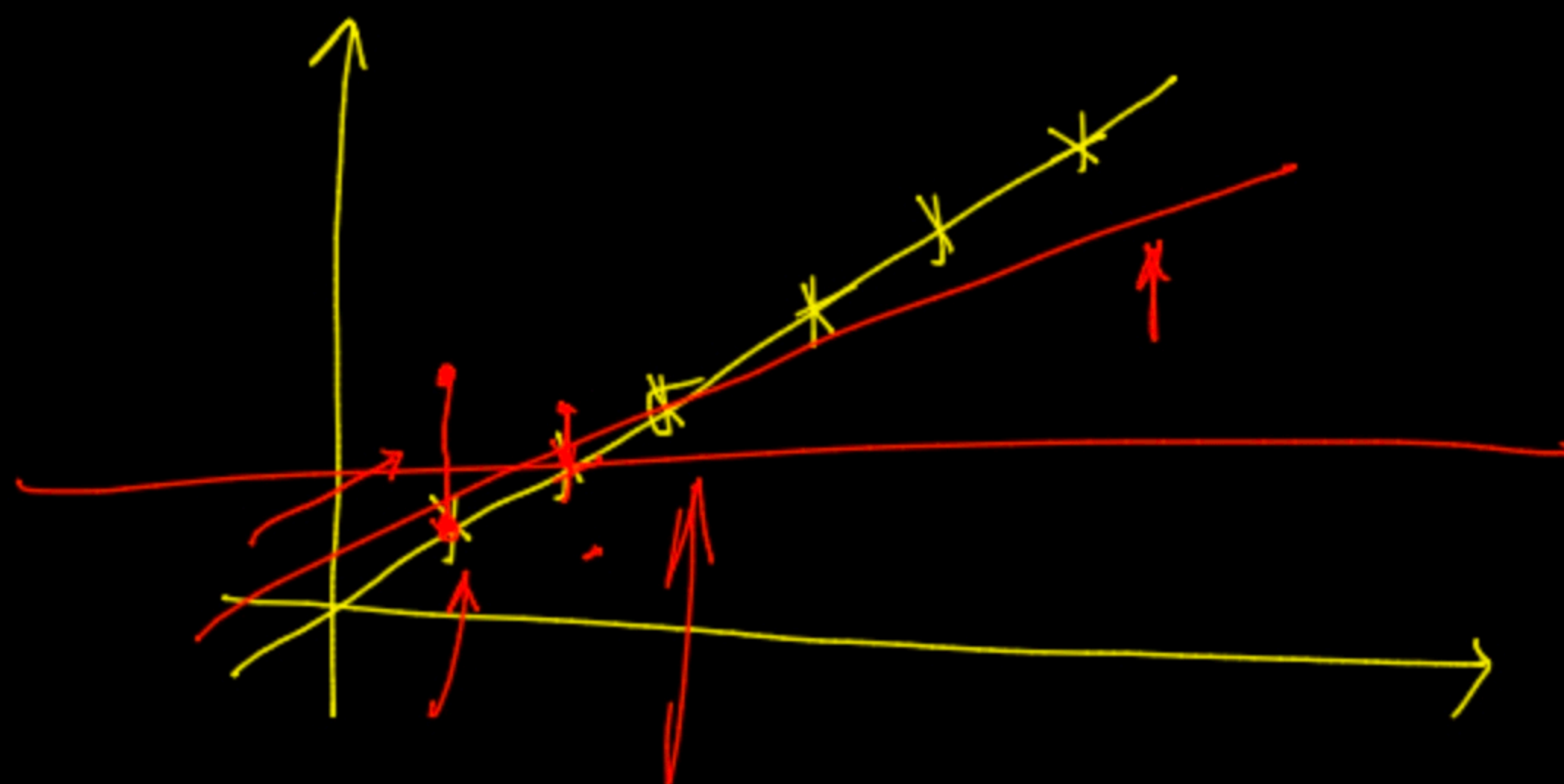
Our objective

↳ find a line/plane that best fit the training data

↑
"minimize data"
error →

(Actual - Prediction)

parameters



error

$$\hat{y} = w_1 * x_1 + w_0$$

X1	Y	$\hat{y}$	
342	340	348	42
	390	372	-
	310	320	-

$$\hat{y} = \underline{0.54} * 342 + 0$$

348

error

$$(w_1, w_0) = \underset{w_1, w_0}{\operatorname{argmin}} \left(\sum_{i=1}^n (y_i - \hat{y})^2 \right) \leftarrow$$

$$= \underset{w_1, w_0}{\operatorname{argmin}} \left(\sum_{i=1}^n \underbrace{y - (w^T x_i + w_0)}_{\substack{\uparrow \\ \text{square-loss}}} \right)^2$$

Regularization $\rightarrow$ $\textcircled{L1}$
 $\textcircled{L2}$

$$\underset{w_1, w_0}{\operatorname{argmin}} \left(\sum_{i=1}^n (y_i - \hat{y})^2 \right) + \lambda \|w\|^2$$

CASES

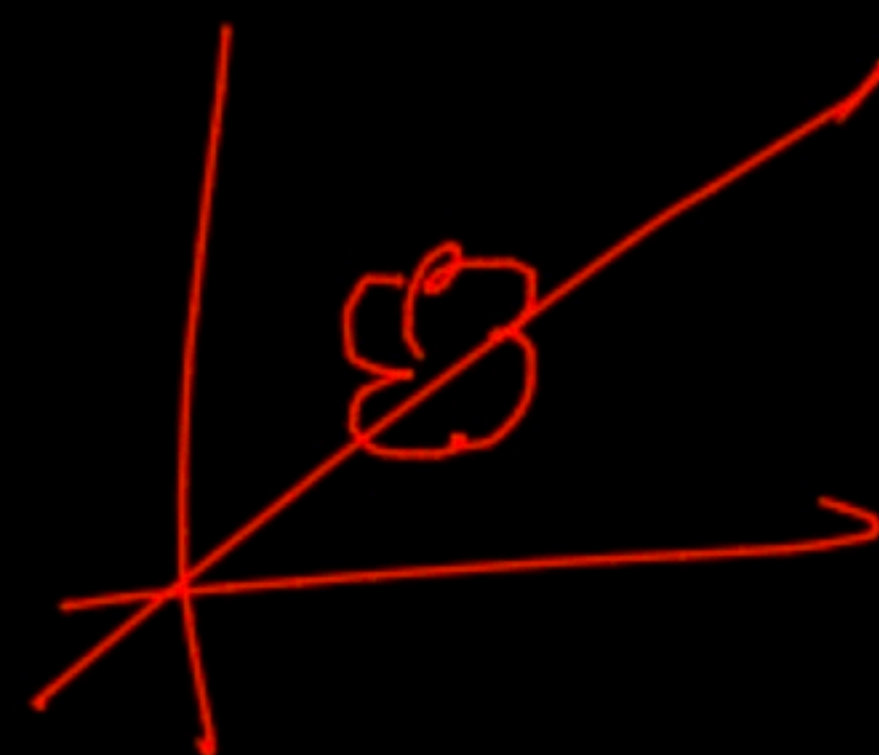
↳ Log Regression → same

↳ Imbalance data →

↳ FE → same

(w1)

↳ (FE / FT)



outlier → X ← (LR)